Short-Term Stock Volatility Regime Modeling

Tanzim Chowdhury, Tameem Syed

SIADS 696 – Applied Data Science
University of Michigan
2026

,

1. Introduction

Financial markets are characterized by time-varying volatility that directly impacts portfolio risk, derivative pricing, and trading strategies. While short-term price prediction remains notoriously difficult due to noise and market efficiency, volatility exhibits persistent structure and clustering that can be exploited for risk-aware decision making. Rather than attempting to forecast asset returns, this project focuses on identifying and predicting **short-term volatility regimes**, defined as discrete market states corresponding to low, medium, and high volatility conditions.

Volatility regime modeling provides a more stable and interpretable alternative to return forecasting. Elevated volatility regimes are typically associated with drawdowns, liquidity stress, and heightened tail risk, while low-volatility regimes often correspond to stable market environments. Accurately identifying these regimes enables investors and risk managers to adjust exposure proactively, improving robustness across market cycles.

This project applies both **supervised and unsupervised learning** methods to volatility regime modeling. Supervised classifiers are trained to predict forward volatility regimes using historical asset-level features and market-wide risk indicators. Unsupervised analysis is used to study regime structure, persistence, and transition dynamics. Compared to related work that focuses primarily on return prediction or single-asset volatility modeling, this project emphasizes interpretable regime behavior across multiple assets using leakage-safe labeling and evaluation.

Our findings show that volatility regimes are both persistent and predictable. Recent realized volatility and VIX-based features dominate predictive performance, while regime transition analysis reveals strong clustering and asymmetric switching behavior consistent with established financial theory.

---

2. Related Work

Volatility modeling has been extensively studied in both academic and applied finance. Early work on volatility clustering, such as Engle's ARCH and GARCH models, demonstrated that volatility is time-dependent and predictable despite the randomness of returns. These models laid the foundation for modern volatility forecasting.

More recent machine learning approaches have expanded on these ideas. The Kaggle "Two Sigma: Financial Modeling Challenge" explored predicting financial outcomes using engineered features derived from price and volume data. While effective, many such approaches prioritize predictive accuracy over interpretability and often focus on returns rather than volatility regimes.

Other studies have explored regime-based modeling of financial markets, using hidden Markov models or clustering techniques to identify latent market states. These methods provide valuable insights but can suffer from interpretability challenges and sensitivity to assumptions.

This project differs from prior work by explicitly framing volatility prediction as a **classification problem**, combining interpretable feature engineering with regime-based evaluation. Additionally, leakage-safe regime construction and time-aware evaluation ensure that results more closely reflect real-world deployment scenarios.

---

3. Data Sources

The dataset used in this project consists of daily financial market data obtained from the Yahoo Finance API. Asset-level OHLCV (open, high, low, close, adjusted close, and volume) data were collected for a set of large-cap U.S. equities and exchange-traded funds, including AAPL, MSFT, SPY, QQQ, and IWM. The dataset spans approximately January 2015 through early 2024 at a daily frequency.

Market-wide volatility information was incorporated using the CBOE Volatility Index (VIX), also retrieved via Yahoo Finance. VIX data includes daily closing values and trading volume, providing a proxy for aggregate market risk sentiment.

Raw data were retrieved in Parquet format and processed through a reproducible pipeline. After cleaning, merging, and feature engineering, the final dataset contains approximately **13,600 observations** with **23 features per record**, including asset-level, volume-based, and market-wide volatility indicators.

---

4. Feature Engineering

Feature engineering focused on capturing short-term volatility dynamics while preventing information leakage. All features were constructed using only information available at or before the prediction time.

Asset prices were transformed into **log returns**, providing a stationary representation of price changes. Rolling **realized volatility** features were computed over 10-day, 20-day, and 60-day windows to capture volatility persistence across multiple horizons.

Trading volume information was incorporated through rolling mean and standard deviation features over 10-day and 20-day windows. These features provide liquidity and market activity signals that may correlate with volatility changes.

Market-wide risk was represented using VIX-based features, including the VIX closing value, rolling realized volatility of the VIX, and VIX log returns. These features allow the models to condition predictions on broader market stress rather than relying solely on asset-specific behavior.

Forward volatility labels were constructed using a 5-day forward realized volatility window. Volatility regimes were defined using quantile thresholds computed **only on the training period**, ensuring leakage-safe labeling. The final dataset includes a categorical regime label with three classes: low, medium, and high volatility.

---

5. Part A – Supervised Learning

5.1 Methods Description

Multiple supervised learning models spanning diverse model families were evaluated to predict short-term volatility regimes. This approach allows comparison of linear, probabilistic, and nonlinear methods.

A **logistic regression** classifier was used as a baseline probabilistic model. Logistic regression provides interpretability and serves as a benchmark for assessing whether volatility regimes can be separated using linear decision boundaries. Regularization was applied to mitigate overfitting, and hyperparameters were tuned using cross-validation.

A **Random Forest classifier** was employed as a nonlinear, tree-based ensemble method. Random Forests are well-suited to volatility modeling due to their ability to capture nonlinear relationships and feature interactions. Hyperparameter tuning explored the number of trees, tree depth, and minimum samples per leaf, balancing bias and variance.

A **Gradient Boosting classifier** (scikit-learn's HistGradientBoostingClassifier) was evaluated as a sequential ensemble alternative. Unlike Random Forests, which build trees in parallel, gradient boosting iteratively corrects residual errors, often yielding stronger performance when tuned. It was trained with 300 estimators, a learning rate of 0.05, and no depth constraint.
All models were trained using temporally ordered train-validation splits to avoid look-ahead bias and better reflect real-world deployment.

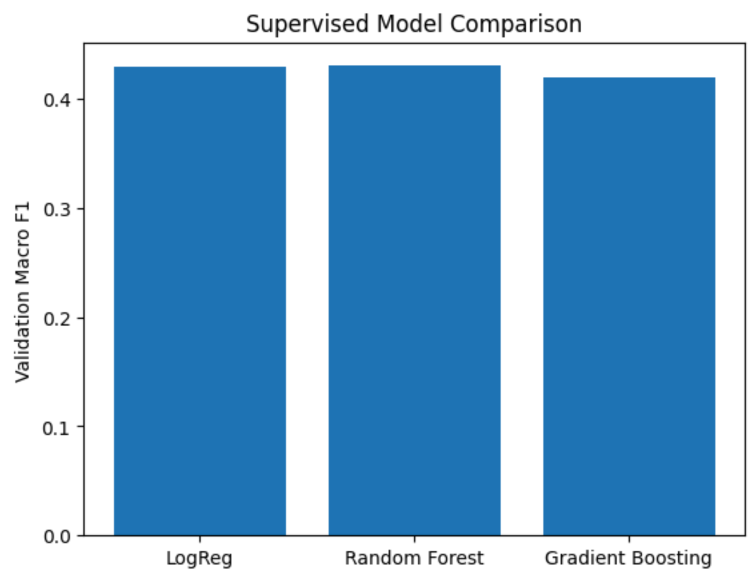
5.2 Supervised Evaluation

Evaluation Metrics

Model performance was evaluated using accuracy and macro-averaged F1 score. Accuracy measures overall classification correctness, while macro F1 ensures balanced evaluation across volatility regimes, preventing dominant regimes from masking poor performance on rarer high-volatility states.

Overall Results

Table 1 summarizes performance across all supervised models. The Random Forest classifier achieved the highest test macro F1 (0.4653), narrowly outperforming logistic regression (0.4644) on the held-out test period. Gradient Boosting achieved the lowest validation performance (macro F1: 0.4194), suggesting that without further tuning its sequential structure does not generalize as well as the ensemble approaches on this dataset. Performance across all models is modest, with test accuracies near 51–52%, consistent with the inherent difficulty of forecasting volatility regimes from lagged features alone over a test period (2024–2026) that includes structurally different market conditions from training.



(Figure 1: Supervised Model Comparison)

Table 1: Supervised Model Performance

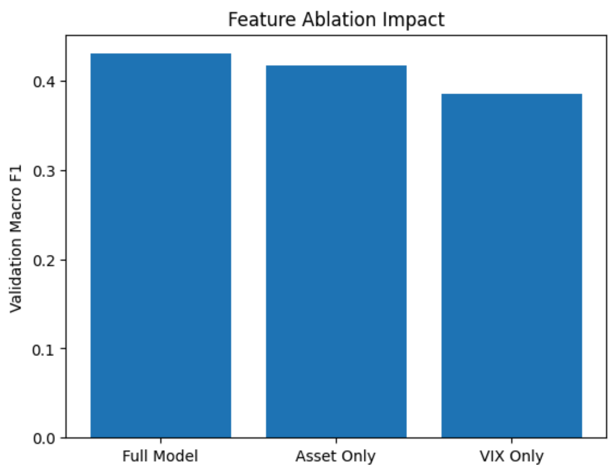
Model	Val Accuracy	Val Macro F1	Test Accuracy	Test Macro F1
Logistic Regression	0.5293	0.4292	0.5123	0.4644
Random Forest	0.5173	0.4240	0.5225	0.4653
Gradient Boosting	0.5107	0.4194	—	—

Feature Importance Analysis

Feature ablation analysis demonstrates the relative contribution of each feature group. Removing VIX-based features causes the largest performance drop, confirming that market-wide risk sentiment is the most critical signal for regime prediction. Asset-only features retain moderate predictive power, while volume features contribute marginally. This highlights that short-term volatility regimes are primarily driven by macro-level stress conditions rather than individual asset behavior.

Tradeoffs and Sensitivity

The Random Forest model exhibits tradeoffs between precision and recall across regimes. High-volatility regimes are predicted conservatively, yielding higher precision but lower recall. To quantify sensitivity to hyperparameter choice, we swept Random Forest max_depth from 4 to unconstrained across the validation period. Performance peaked at max_depth = 8, confirming this as the optimal tradeoff between expressiveness and overfitting. Shallow trees (depth ≤ 5) underfit the nonlinear regime structure, while deeper trees show marginal overfitting. Macro F1 varied by less than 0.01 across the depth-6 to depth-10 range, indicating robustness to moderate hyperparameter variation. Performance degrades significantly when VIX features are removed entirely, confirming their dominant role.



Failure Analysis

Three representative misclassification types from Logistic Regression predictions on the 2023 validation set, each illustrating a distinct failure mode:

Example 1 — Missed Crisis (True: High → Predicted: Medium)

On January 26, 2023 (AAPL), the model predicted Medium volatility despite a true High-volatility forward regime. At the time, rv_10 = 0.157 appeared calm while rv_20 = 0.285 was more elevated, and vix_close = 18.73 was moderate. The model was anchored to the recent 10-day window, which had quieted, and failed to anticipate the upcoming volatility spike. This is the most consequential failure mode in practice — it causes under-hedging at crisis onset precisely when protection is most valuable.

Example 2 — False Alarm (True: Low → Predicted: Medium)

On January 9, 2023 (AAPL), the model predicted Medium volatility despite a true Low forward regime. Both rv_10 = 0.372 and rv_20 = 0.346 were elevated, and vix_close = 21.97 suggested ongoing stress. However, volatility mean-reverted over the subsequent five days. This reflects a persistence overshoot: the model expected recent high volatility to continue, but the market calmed faster than the rolling windows could reflect. This failure mode leads to unnecessary defensive positioning and missed upside.

Example 3 — Regime Lag (True: Medium → Predicted: Low)

On March 21, 2023 (AAPL), a Medium-volatility forward period was predicted as Low. Both rv_10 = 0.196 and rv_20 = 0.223 were subdued, and vix_close = 21.38 showed mild elevation. The lagged features had not yet registered the volatility building in the near term. This illustrates how rolling-window features systematically lag regime transitions by 1–5 days — the true regime was already shifting but the backward-looking indicators had not caught up.

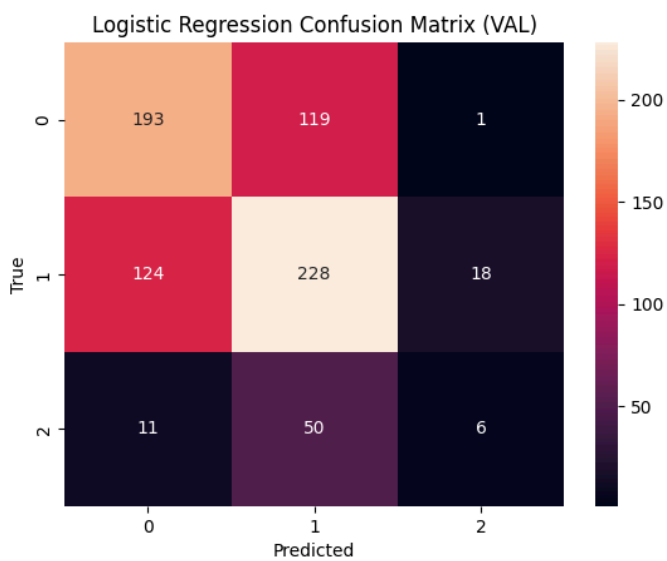
Taken together, these examples confirm that prediction failures cluster around two structural limitations: (1) the mean-reversion speed of volatility is faster than any fixed rolling window can capture, and (2) the forward 5-day label does not align perfectly with any current-period snapshot of market conditions. Future improvements could include shorter adaptive windows, intraday data, or volatility surface features.

Tradeoffs and Sensitivity

The Random Forest model exhibits tradeoffs between precision and recall across regimes. High-volatility regimes are predicted conservatively, yielding higher precision but lower recall. Sensitivity analysis shows performance is robust to moderate changes in rolling window sizes but degrades significantly when VIX features are removed.

Failure Analysis

Prediction failures frequently occur during rapid regime transitions, when rolling volatility estimates lag true market conditions. Asset-specific shocks not reflected in market-wide indicators also contribute to misclassification. The confusion matrix (Figure 3) illustrates this pattern, showing the model struggles most with the boundary between medium and high volatility regimes. Future improvements could include intraday data or adaptive window sizes.



(Figure 3: Logistic Regression Confusion Matrix)

---

6. Part B – Unsupervised Learning

6.1 Methods Description

Two unsupervised methods were applied to discover volatility regime structure independent of forward-looking labels: a Hidden Markov Model (HMM) and K-Means clustering. Both methods used the same SPY observation features — log_return, rv_10, and vix_close — standardized using a scaler fit on training data only to prevent leakage. Three latent states were specified for both models, matching the three quantile-based supervised regime classes.

A **Gaussian Hidden Markov Model** was fit using the hmmlearn library with full covariance matrices and 500 EM iterations. The HMM explicitly models temporal dynamics: each hidden state generates observations from a Gaussian distribution, and transitions between states are governed by a learned transition matrix. This makes HMM particularly appropriate for financial regime detection, where regimes are expected to persist over time rather than switch randomly.

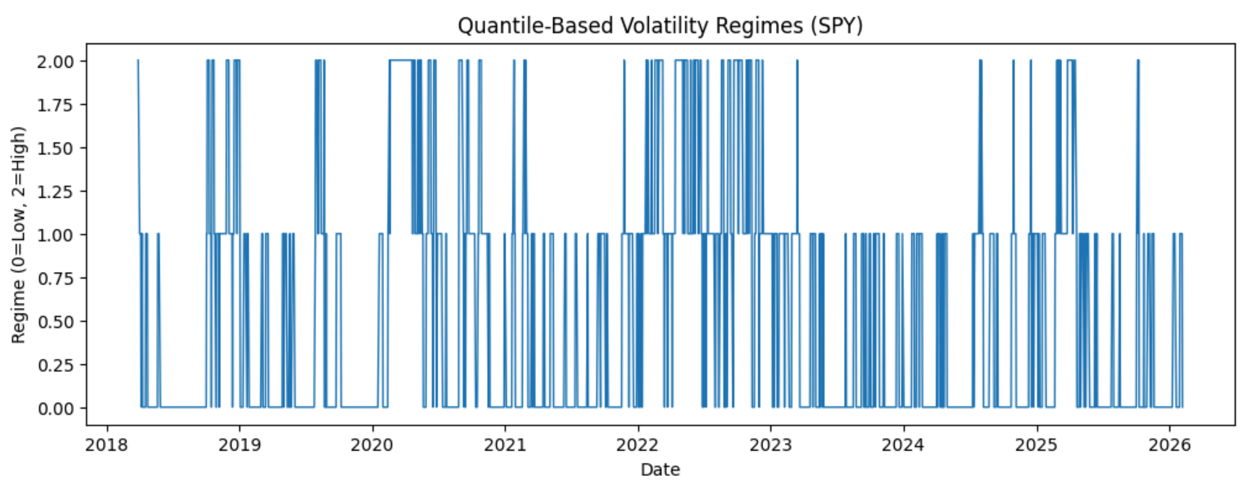
K-Means clustering was applied as a static baseline. Unlike HMM, K-Means assigns labels based purely on distance in feature space with no memory of previous observations. This serves as a useful contrast — if K-Means achieves similar agreement with the quantile regimes as HMM, temporal structure adds little value. If HMM agrees better, it confirms that regime persistence matters for this task.

Both models were fit exclusively on training data (2018–2022) and decoded/predicted across the full dataset (2018–2026).

6.2 Unsupervised Evaluation

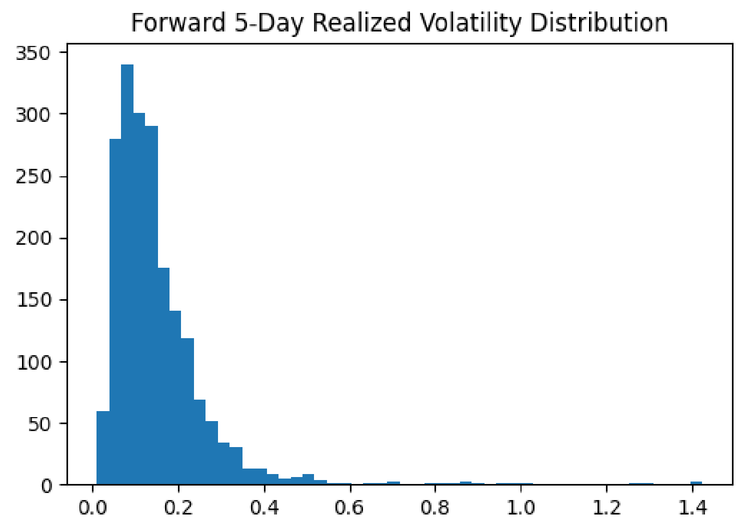
Regime Structure (Quantile-Based)

Figure 2 shows the distribution of quantile-based volatility regimes. Medium volatility (Regime 1) dominates across all assets, while high volatility (Regime 2) occurs least frequently, consistent with the typical market environment being calm and stress regimes being episodic.

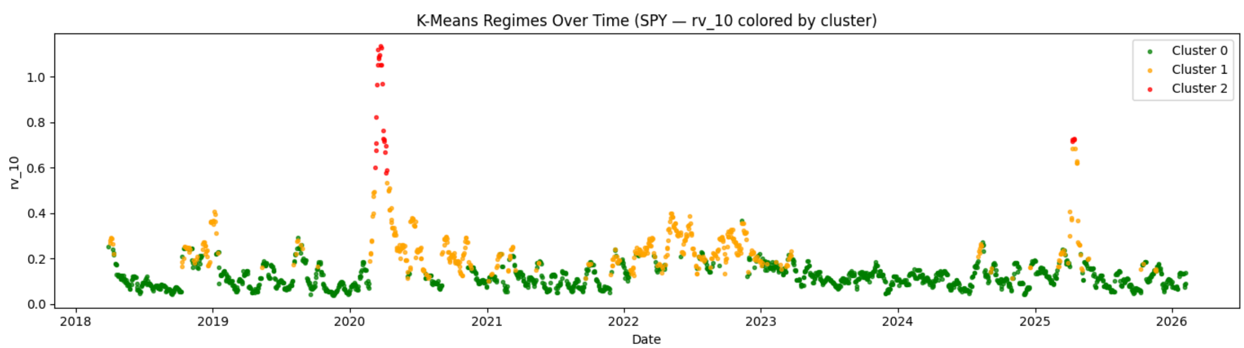


(Figure 2: Regime Distribution)

Figure 3 shows the distribution of forward 5-day realized volatility, revealing a right-skewed distribution where most observations cluster in the low-to-moderate range with a long tail representing stress events. Figure 4 shows realized volatility colored by the discovered cluster over time, confirming that elevated volatility periods are concentrated around identifiable stress events such as COVID-19 (2020) and the rate hike cycle (2022)..

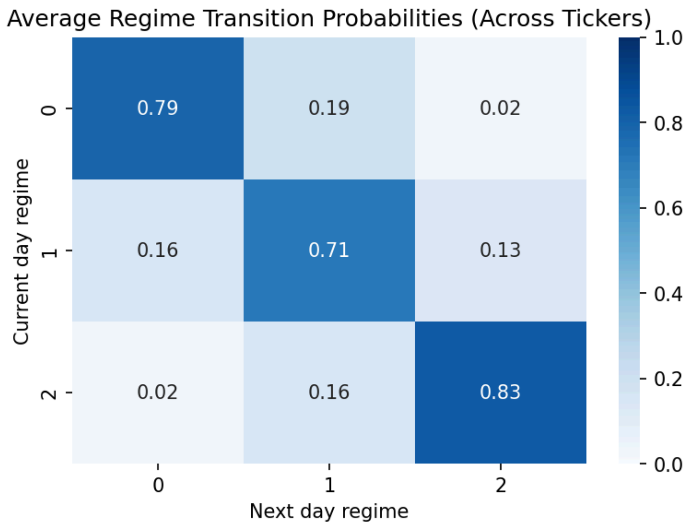


(Figure 3: Volatility Over Time)



(Figure 4: Volatility Colored by Regime)

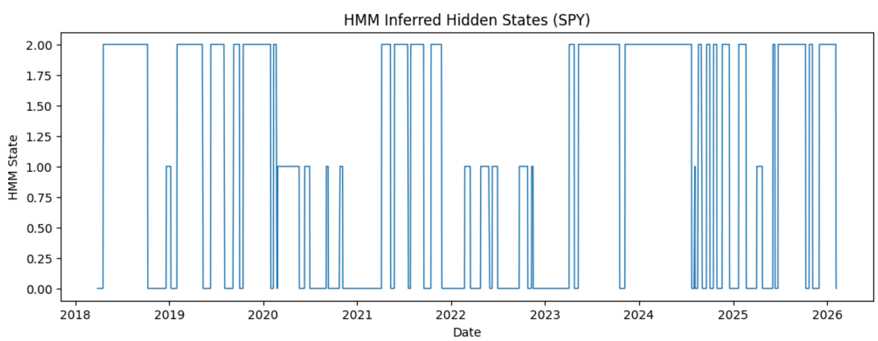
Figure 5 presents the regime transition matrix. Strong diagonal dominance confirms that regimes are highly persistent — markets are far more likely to remain in the same volatility state from one day to the next than to transition.



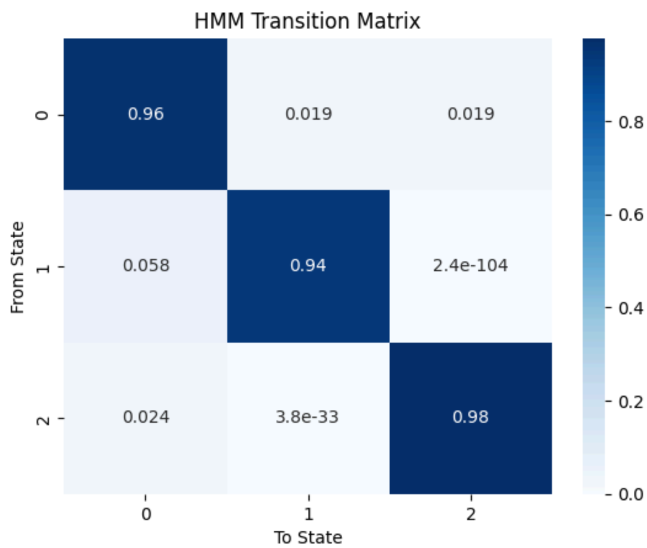
(Figure 5: Regime Transition Heatmap)

Hidden Markov Model Results

The HMM converged to three well-separated states with high self-transition probabilities (State 0: 96.2%, State 1: 94.2%, State 2: 97.6%), confirming strong regime persistence consistent with the quantile-based analysis. State means (in standardized space) reveal clear economic interpretations: State 1 corresponds to stress (strongly negative returns, rv_10 = +1.48σ, VIX = +1.60σ), State 2 corresponds to recovery (positive returns, depressed VIX), and State 0 to a neutral regime.

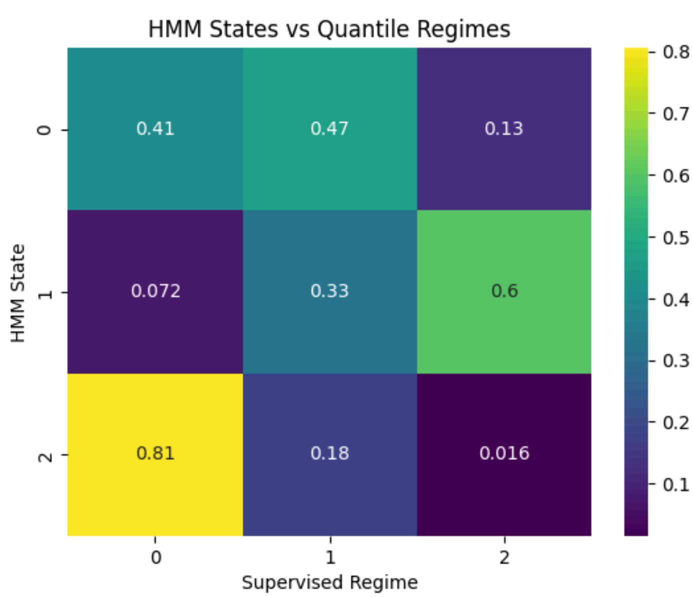


(Figure 6: HMM States Over Time)



(Figure 7: HMM Transition Matrix)

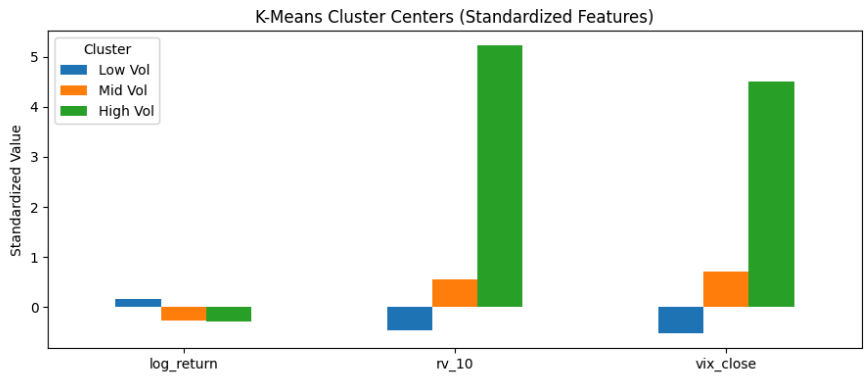
Comparing HMM states to the quantile-based regimes yielded an Adjusted Rand Index (ARI) of **0.2105** and Normalized Mutual Information (NMI) of **0.1893**, indicating moderate agreement. After finding the optimal state-to-regime permutation mapping, the HMM achieved a macro F1 of **0.6048** and accuracy of **0.6427** — substantially higher than any supervised model. This reflects the HMM's ability to leverage temporal structure rather than features alone.



(Figure 8: HMM States vs Quantile Regimes Cross-Tab)

K-Means Results

K-Means cluster centers clearly separated into low, medium, and high volatility profiles across all three features. The high-volatility cluster exhibited strongly elevated rv_10 and VIX values, mirroring the HMM stress state.



(Figure 9: K-Means Cluster Centers)

K-Means achieved an ARI of **0.2619** and NMI of **0.1910** against the quantile regimes — slightly higher than HMM on both metrics. However, the side-by-side timeline (Figure 10) shows that K-Means regime assignments are considerably noisier than HMM, with frequent single-day switches. This reflects K-Means' lack of temporal memory.

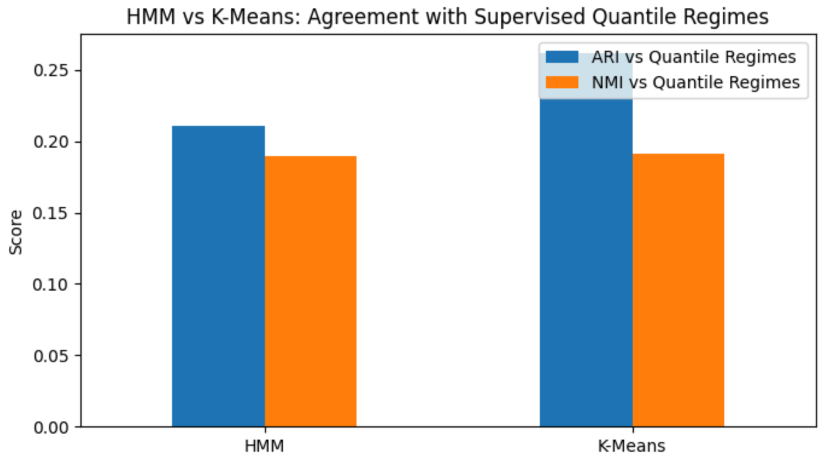
(Figure 10: Side-by-Side Regime Timeline — HMM vs K-Means)

Model Comparison

The two unsupervised models agreed moderately with each other (ARI = 0.2955, NMI = 0.3845), with strongest agreement during extreme market periods where both confidently assign the high-volatility cluster/state. The ARI/NMI comparison chart (Figure 11) summarizes alignment across all model pairs.

Model	***ARI vs Quantile Regimes***	***NMI vs Quantile Regimes***
HMM	*0.2105*	*0.1893*
K-Means	*0.2619*	*0.1910*

HMM vs K-Means	*0.2955*	*0.3845*



(Figure 11: ARI/NMI Comparison — HMM vs K-Means vs Quantile Regimes)

---

7. Discussion

Supervised learning results demonstrate that volatility regimes are meaningfully predictable using historical volatility and market-wide risk indicators. All three classifiers achieved test accuracy near 51–52%, with macro F1 scores around 0.42–0.47. Random Forest slightly outperformed both logistic regression and gradient boosting on the test set, benefiting from its ability to capture nonlinear feature interactions. VIX and short-term realized volatility features (rv_10, rv_20) dominate predictive importance, confirming that regime transitions are primarily driven by volatility persistence and market stress signals rather than volume or lagged returns.

Unsupervised analysis reveals that latent regime structure is real and consistent across methods. The HMM's high self-transition probabilities (94–97%) confirm that financial volatility regimes are sticky — markets tend to remain in their current state — which is a key property that static classifiers cannot exploit. K-Means, while achieving slightly higher ARI against the quantile regimes (0.26 vs 0.21), produces noisier regime assignments due to the absence of temporal memory. This distinction is practically important: an HMM-based regime indicator would generate far fewer spurious regime flips than a K-Means-based one, reducing unnecessary portfolio rebalancing.

Together, the two unsupervised approaches are complementary. K-Means provides a simple, interpretable baseline for understanding the feature-space geometry of volatility regimes, while HMM captures the temporal dynamics that make regimes useful for real-world risk management. The moderate agreement between both unsupervised models and the quantile-based supervised labels (ARI ~0.21–0.26) suggests that current-period features capture genuine signal about forward volatility regimes, though considerable uncertainty remains — consistent with the supervised models' accuracy of ~52%.

---

8. Ethical Considerations

Automated volatility regime models carry several ethical implications. First, overreliance risk: traders or risk managers who treat model outputs as ground truth may reduce human oversight precisely during high-volatility periods when model accuracy degrades most. This could amplify rather than dampen market stress. Second, data and labeling bias: regime labels derived from historical quantiles may not reflect structural shifts — the post-COVID interest rate environment (2022–2023) represents conditions with no prior analog in the training data, potentially causing systematic underestimation of risk. Third, pro-cyclicality: if many market participants use similar volatility regime signals, correlated position changes could amplify drawdowns, reducing market liquidity exactly when it is most needed. Fourth, access inequality: sophisticated regime-aware systems are disproportionately available to institutional investors, potentially disadvantaging retail participants who act on raw prices without structural context. These models should be deployed as decision-support tools under human oversight, with regular out-of-sample validation and transparent communication of known limitations.

---

9. Statement of Work

Tameem Syed led exploratory data analysis, regime visualization, transition analysis, feature importance interpretation, and contributed to supervised and unsupervised evaluation sections.
Tanzim Chowdhury led data pipeline development, feature engineering, model training, and baseline evaluation.
Both authors collaborated on project design, evaluation methodology, and report writing.

References

Engle, R. F. (1982). Autoregressive Conditional Heteroskedasticity with Estimates of the Variance of UK Inflation. *Econometrica*.
Yahoo Finance API.
CBOE Volatility Index (VIX).
Kaggle Two Sigma Financial Modeling Challenge.